Project Overview

Haptic Speak is an innovative capstone project in which auditory speech is transformed into tactile feedback through a wearable device. Advanced sensor technology and real-time signal processing are employed so that spoken language is captured and converted into distinct haptic patterns. The challenges faced by individuals with hearing impairments are addressed and a novel communication medium is pioneered.

A compact array of microphones, sophisticated processing algorithms, and multiple haptic actuators are integrated into the system to ensure that audio input is converted into tactile stimuli perceivable through the skin. Essential speech elements such as rhythm, intonation, and emphasis are conveyed through these haptic patterns. A new tactile "language" is introduced to be learned and recognized by the deaf community.

Significant contributions are made by bridging auditory and tactile modalities in the fields of human-computer interaction and assistive technology. Wide-ranging applications are anticipated, ranging from enhancements in everyday communication for the hearing-impaired to the provision of alternative sensory experiences in noise-intensive environments. Ultimately, the potential of haptic feedback to redefine the experience and understanding of spoken language is underscored.

Need Analysis

The need for the proposed project is established by the challenges that are encountered in the realm of communication by individuals with hearing impairments and by the limitations that are observed in existing assistive technologies. It is recognized that traditional methods of communication are predominantly reliant on auditory cues, which consequently exclude a significant portion of the population from fully participating in social, educational, and professional environments. Existing systems that convert sensory information have been developed, yet their capacity to accurately and efficiently convey the intricate nuances of spoken language has been found to be insufficient.

Real-world scenarios, such as high-noise environments and emergency situations, are confronted with difficulties that restrict the use of conventional auditory-based communication systems. A need is thereby identified for a solution that can reliably translate the complex features of speech such as rhythm, intonation, and emphasis into a modality that is universally perceivable. The integration of haptic feedback with speech recognition technology is expected to offer a transformative approach, as demonstrated by previous research in related fields. It is anticipated that this new method will not only enhance the quality of life for the hearing-impaired but also provide broader insights into multimodal communication systems.

The societal relevance of the proposed project is underscored by its potential to bridge communication gaps and to foster inclusivity in various settings. Widespread adoption of this technology is projected to be facilitated by its modular design and adaptability to multiple real-world applications. Future enhancements are expected to be guided by ongoing research and practical implementations, thereby ensuring that the system remains aligned with evolving communication challenges and technological advancements.

Literature Survey

Advancements in haptic technology and real-time speech processing have been greatly enhanced by recent innovations [2], [4], [5], and [8], thereby increasing the potential for alternative communication systems for the deaf [1]. A literature survey was conducted for Haptic Speak so that relevant technologies, trends, and research gaps in converting ambient speech into tactile language could be explored. The need for intuitive and learnable haptic languages that can effectively substitute auditory information was emphasized by research on wearable haptic systems [10]. The benefits of sensor-based and wearable devices in enhancing communication were highlighted by studies on speech-to-haptic conversion and tactile language acquisition

[6], while challenges related to user training, device ergonomics, and integration with real-time speech recognition were also revealed [7]. Based on these findings, best practices for dynamic haptic language mapping are intended to be implemented by Haptic Speak so that a user-friendly, comfortable system capable of providing real-time, accurate tactile feedback can be realized [9].

Table 3.1: Literature Survey

S.No	Year	Title	Key Findings	Research Gap	Citations
1	2019	Haptic Communication of Language	Historical overview and applications of tactile language.	No standardized haptic language for wearables.	T. Ifukube and R. L. White, "A Speech Processor with Lateral Inhibition for an Eight Channel Cochlear Implant and its Evaluation," in IEEE 1987.
2	2020	SmartFingerBraille: A Tactile Sensing and Actuation Glove for Deafblind People	Tactile glove demonstrated Braille-based language transmission.	Limited scalability and vocabulary need adaptation.	L. Chai, J. Du, Q.-F. Liu, and C.-H. Lee, "A Cross-Entropy-Guided Measure (CEGM) for Assessing Speech Recognition Performance and Optimizing DNN-Based Speech Enhancement," in IEEE/ACM 2021.
3	2019	GLOS: GLOve for Speech Recognition	Wearable converted speech to haptic communication.	Haptic learning curve and vocabulary lacked depth.	Y. Ma, C. Zhang, Q. Chen, W. Wang, and B. Ma, "Tuning Large Language Model for Speech Recognition With Mixed-Scale Re-Tokenization," in IEEE 2024.
4	2020	Acquisition of 500 English Words through a Tactile Phonemic Sleeve (TAPS)	Wearable sleeves aided language learning with haptics.	Tactile language retention and efficacy under-researched.	A. Kumar, S. Patel, and D. Lee, "Real-Time Noise Suppression for Speech Enhancement Using Adaptive Filtering Techniques," in IEEE Transactions 2019.

Theory observed with problem area

3.1 Integrated Systems:

It was revealed by existing studies that speech recognition is not fully integrated with haptic translation. Multi-sensor data and real-time voice processing are intended to be combined by Haptic Speak so that personalized, adaptive haptic feedback is generated.

3.2 Ergonomic and Scalable Design:

It has been observed that several wearable systems cause discomfort or are limited by static designs. A comfortable and user-friendly device (e.g., a wristband or glove) is intended to be developed by Haptic Speak, so that continuous wear without irritation is enabled.

3.3 Real-Time Conversion and Accuracy:

The low latency required for effective communication and the accuracy of haptic translations have not been achieved by many current systems. A focus on minimizing latency (targeted to be below 100 ms) and improving the precision of haptic cues through advanced signal processing techniques is intended by Haptic Speak.

3.4 Intuitive Haptic Language Mapping:

A significant research gap in the creation of an intuitive, learnable haptic language has been identified. Standardized mappings for key speech features (such as pitch, rhythm, and amplitude) into distinct tactile patterns are intended to be established by Haptic Speak.

3.5 User Training and Adaptability:

It has been found that the learning curve associated with interpreting haptic signals remains a challenge. User-centric design and iterative feedback from the deaf community are intended to be incorporated by Haptic Speak so that training protocols are refined and effective communication is ensured.

Objectives

- A wearable device that converts spoken speech into distinct haptic patterns is to be developed.
- A new haptic language, like Braille for the blind or sign language for the deaf, is to be created for effective communication.
- The rendering of essential speech features such as rhythm, intonation, and stress into tactile stimuli is to be optimized.
- The usability and performance of the device are to be validated through systematic user testing and iterative design refinements.

Methodology

We will be needing 1-2 weeks to complete this methodology section as we were assigned the mentor very recently.

Work Plan

A short work plan for the academic year 2025 is given below as on how we are going to approach the project.

Table 5.1: Work Plan

Sr. No	Activity	January	February	March	April	May	June	July	August	September	October
1	Group Formation and Problem Identification	■									
2	Literature Review & Finalize Project Proposal (Obtain Approval)		■								
3	Requirement Gathering and System Design			■	■						
4	Hardware & Software Selection (Microphones, Actuators, Processing Units, etc.)				■	■					
5	Develop Initial Speech-to-Haptic Conversion Algorithm					■	■				
6	Prototype Development (Hardware Integration & Software Implementation)						■	■			
7	Testing & Refinement of Haptic Feedback Patterns							■	■		
8	Conduct User Testing & Collect Feedback (Hearing-Impaired Participants)								■	■	
9	Optimize System for Real-Time Processing and Low Latency									■	■
10	Prepare Documentation (Technical Report & Research Paper)										■
11	Finalize Deployment-Ready Prototype, Troubleshooting										■

Project Outcomes

- A fully functional, wearable device has been developed that captures ambient speech by employing a high-quality MEMS microphone and processes the signal in real time via an embedded DSP (e.g., Texas Instruments TMS320C6748). Key speech features, such as pitch, rhythm, and amplitude, are translated into distinct haptic patterns that are delivered through integrated actuators, while the overall system latency is maintained below 100 ms.

- A systematically designed haptic language has been established, whereby speech characteristics are converted into tactile cues that are both intuitive and learnable by deaf users. Iterative user testing and feedback from deaf individuals, audiologists, and speech therapists have been utilized to validate the effectiveness and usability of the haptic communication approach.

- Seamless integration of audio capture, digital signal processing, and haptic actuation modules has been achieved in a compact, ergonomic form factor (e.g., wristband or glove). Optimized software algorithms, including real-time noise cancellation, MFCC extraction, and adaptive filtering, have been implemented to ensure reliable performance under varying ambient conditions.

- A novel assistive technology has been created that provides an alternative communication channel, thereby significantly improving accessibility and the quality of life for individuals with hearing impairments. The device has been designed to serve as both a communication aid and a training tool, enabling users to acquire and refine their understanding of the haptic language.

- Energy efficiency has been realized by achieving a battery life of at least 8 hours, rendering the device suitable for extended daily use. A design that meets regulatory and safety standards has been implemented, thereby paving the way for potential commercialization and wider adoption.

Individual Roles

In this section we are specifying the individual roles and responsibilities of each member.

Table 7.1: Individual Roles & Responsibilities

Name	Role	Responsibility
Raghav Sudan (ENC)	Hardware & Embedded Systems	The wearable device, including microcontrollers and haptic actuators, is being designed and integrated, ensuring seamless communication between hardware components and software processing units.
Suryanshu Anand (ENC)	AI/ML & ASR	Machine learning models for speech recognition and feature extraction are developed, and algorithms are implemented to map speech features (intonation, rhythm, stress) into haptic patterns.
Deepankit Chopra (COE)	Haptic Language Creation	A structured haptic language for representing speech features through tactile stimuli is developed, and haptic feedback patterns are optimized to enhance user perception and communication.
Namrah Sharma (COPC)	Dataset Organization for ML Models	Speech datasets for model training and validation are collected, processed, and organized, while data quality, labeling, and preprocessing are ensured to improve AI model accuracy.
Manu Dev Tyagi (COE)	Software Development	Real-time speech-to-haptic architecture developed.

Course Subjects

In this segment we have identified the course subjects which are relevant in this project.

Table 8.1: Course Subjects

Subject Name	Subject Code	Branch
Embedded Systems	UEC513	ENC
Digital Signal Processing	UEC502	ENC
Analog and Digital Communication Systems	UEC519	ENC
Conversational AI: Accelerated Data Science	UCS551	ENC/COE
Software Engineering	UCS503	COE
Artificial Intelligence	UCS411	COE

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